

backend/ Directory Guide

Python Web Server, AI Engines & Core Business Logic

1. Folder Purpose & Architectural Role

The **backend/** directory is the central brain of CivicPulse-BRICS. It houses the FastAPI/Starlette web server, SQLite database ORM controllers, Google Gemini AI synthesis engines, geopolitical metadata stores, and real-time SMS/WhatsApp messaging gateways.

2. File Breakdown & Dependency Matrix

File / Subdirectory	Detailed Description & Architectural Role
• <code>server.py</code>	Main ASGI web server defining all REST API routes (<code>/api/complaints</code> , <code>/api/demands</code> , <code>/api/brics</code> , <code>/api/auth</code>), static mounts, file upload handlers, and HTML page routes.
• <code>api_client.py</code>	Core database access layer managing SQLite read/write operations (CRUD) for complaints, user accounts, SLA tracking, and engineer dispatch.
• <code>app.py</code>	Development runner script that imports the Starlette ASGI application from <code>server.py</code> and launches Uvicorn locally.
• <code>brics_context.py</code>	Geopolitical data store defining metadata for all 11 BRICS+ nations (capitals, currencies, languages, urban centers, and sectors).
• <code>brics_proposals_engine.py</code>	Manages municipal CapEx proposals and cross-border BRICS joint venture project requests.
• <code>demands_engine.py</code>	Manages community demand proposals submitted by citizens and voter tracking.
• <code>megaplan_engine.py</code>	AI-powered urban planning engine that synthesizes multi-sector MegaPlans from complaint clusters using Google Gemini.
• <code>budget_engine.py</code>	Calculates cost estimations and budget allocation breakdowns for CapEx proposals.
• <code>gemini_helper.py</code>	Centralized wrapper for Google Gemini AI API with fallback handling, structured JSON parsing, and severity triage.
• <code>messaging_gateway.py</code>	Handles outbound multi-channel messaging (WhatsApp, SMS, push alerts) on complaint creation and status resolution.
• <code>sms_gateway.py</code>	Dedicated SMS log manager and Twilio integration module storing SMS audit logs in SQLite.
• <code>tts_helper.py</code>	Text-to-Speech audio synthesis helper for voice accessibility.
• <code>public_data_helper.py</code>	Fetches live environmental context (weather, air quality) from open APIs.
• <code>dpg_spec.py</code>	Digital Public Good specification generator outputting OpenAPI 3.0 specs.

3. Operational Notes & Usage Instructions

Executed directly by Docker container via `uvicorn backend.server:app --host 0.0.0.0 --port 8080`.